

Power and Spares

Master Control

File: power.kicad_sch

File: bcontroller.kicad_sch

PCI Bus

ATA Interface

File: pcibus.kicad_sch

File: IDE.kicad_sch

Agnus Multiadapter

HID and User Ports

File: Agnus-Multi.kicad_sch

File: hid.kicad_sch

Chip RAM Controller

Audio

File: cpu.kicad_sch

File: audio.kicad_sch

PCI Bridge

Video

File: pci-bridge.kicad_sch

File: video.kicad_sch

Motorola Bus Logic

Chip Memory

File: motorola_bus_logic.kicad_sch

File: chip_memory.kicad_sch

Serial

Clocks, ROM, USB

File: serial.kicad_sch

File: clocks.kicad_sch

Floppy and Parallel

PCI Bus Logic

File: floppy.kicad_sch

File: pci_bus_logic.kicad_sch

100 = PCI BUS
200 = CHIPSET
300 = VIDEO
500 = AUDIO
600 = POWER
700 = CHIPSET RAM
800 = VOLTAGE SHIFTER
900 = ATA

BUDDY

DESIGNED WITH KICAD REV 8

Sheet: /
File: AmigaPCI Mainboard.kicad_sch

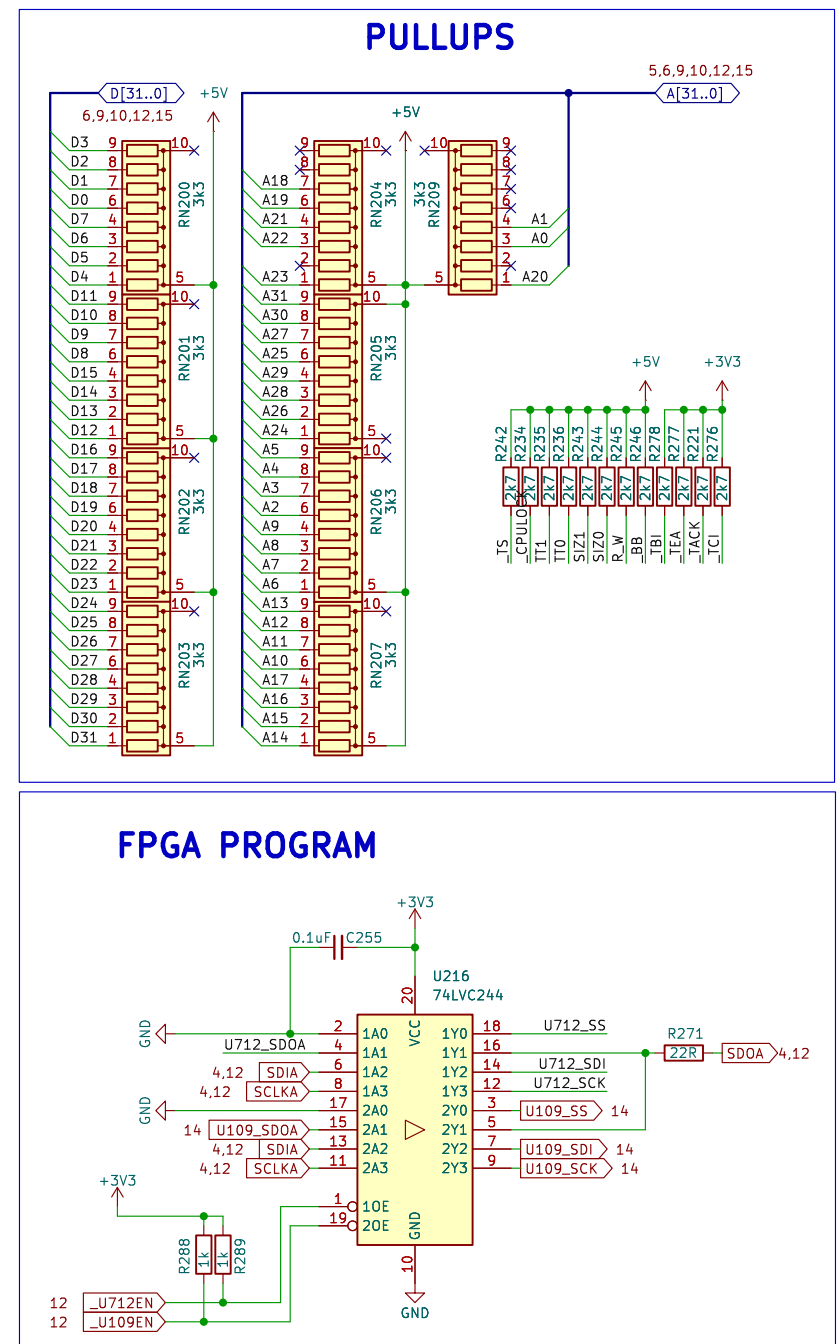
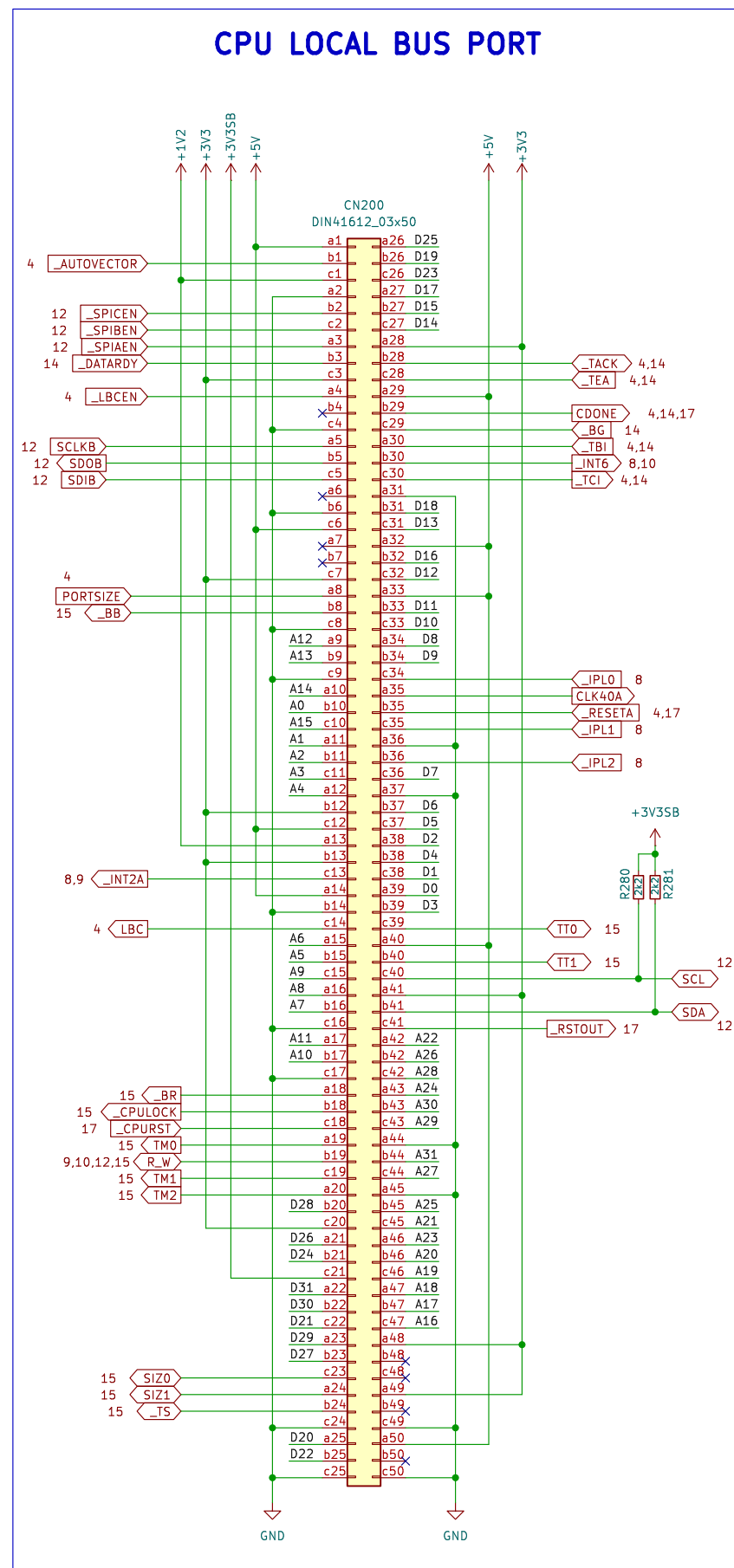
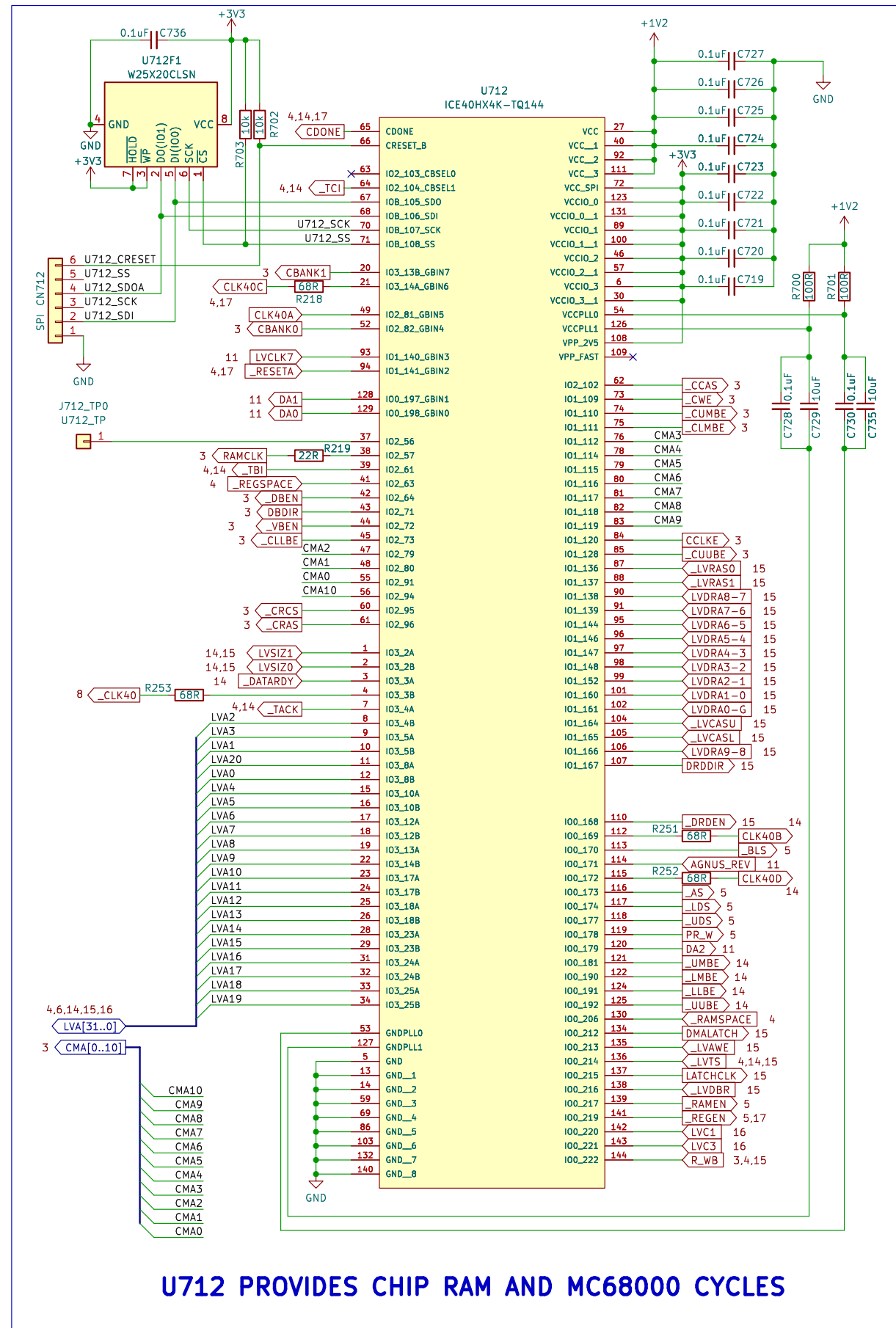
Title: AMIGA PCI

Size: USLedger	Date: 2026-04-13
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Rev: 7.0

Size: 05Ledge1	
KiCad E.D.A. 8.0.9	

id: 1/17



CHIP RAM CONTROLLER

Sheet: /Chip RAM Controller/
File: cpu.kicad_sch

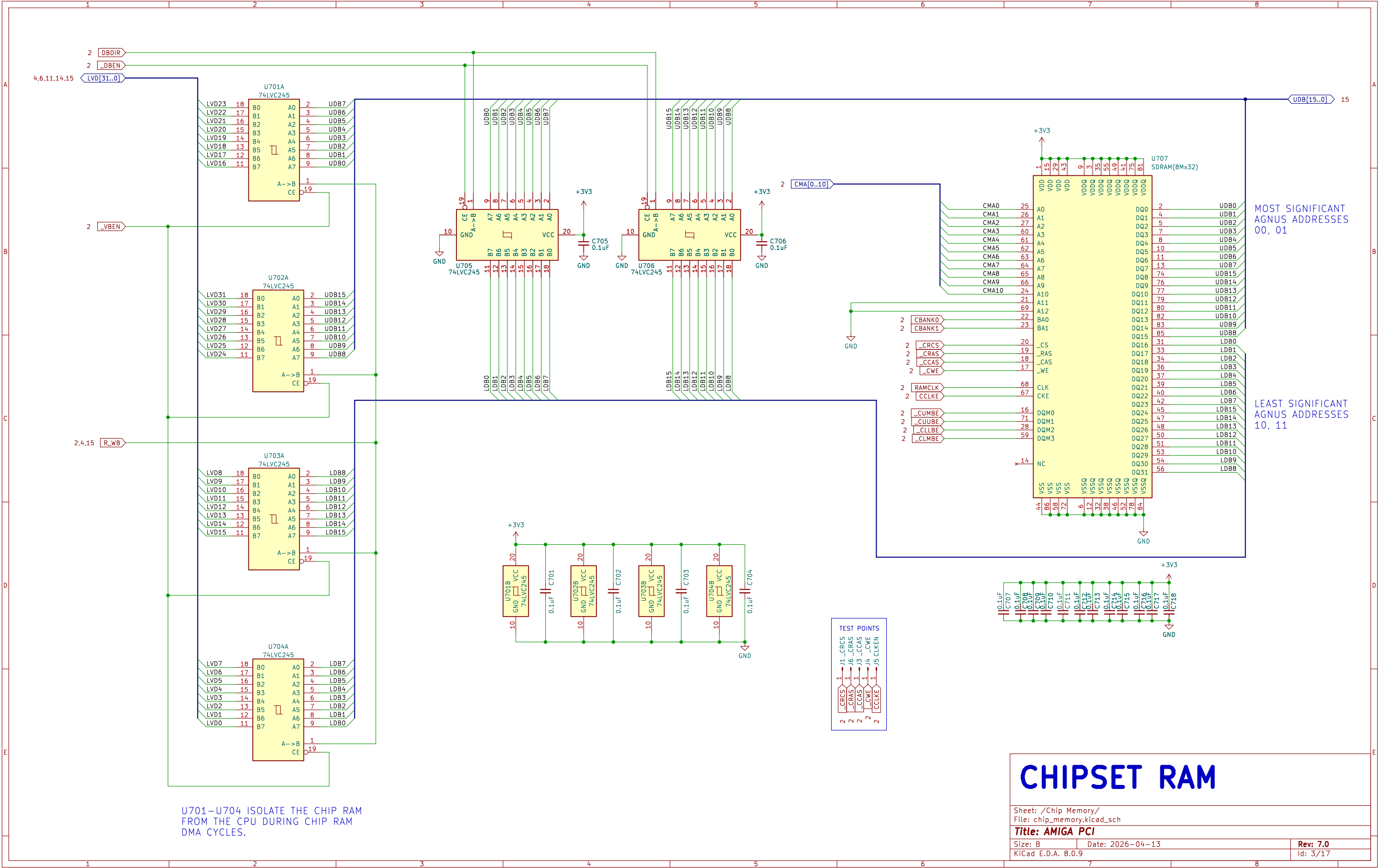
Title: AMIGA PCI

Size: B	Date: 2026-04-13
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KiCad E.D.A. 8.0.9

Rev: 7.0

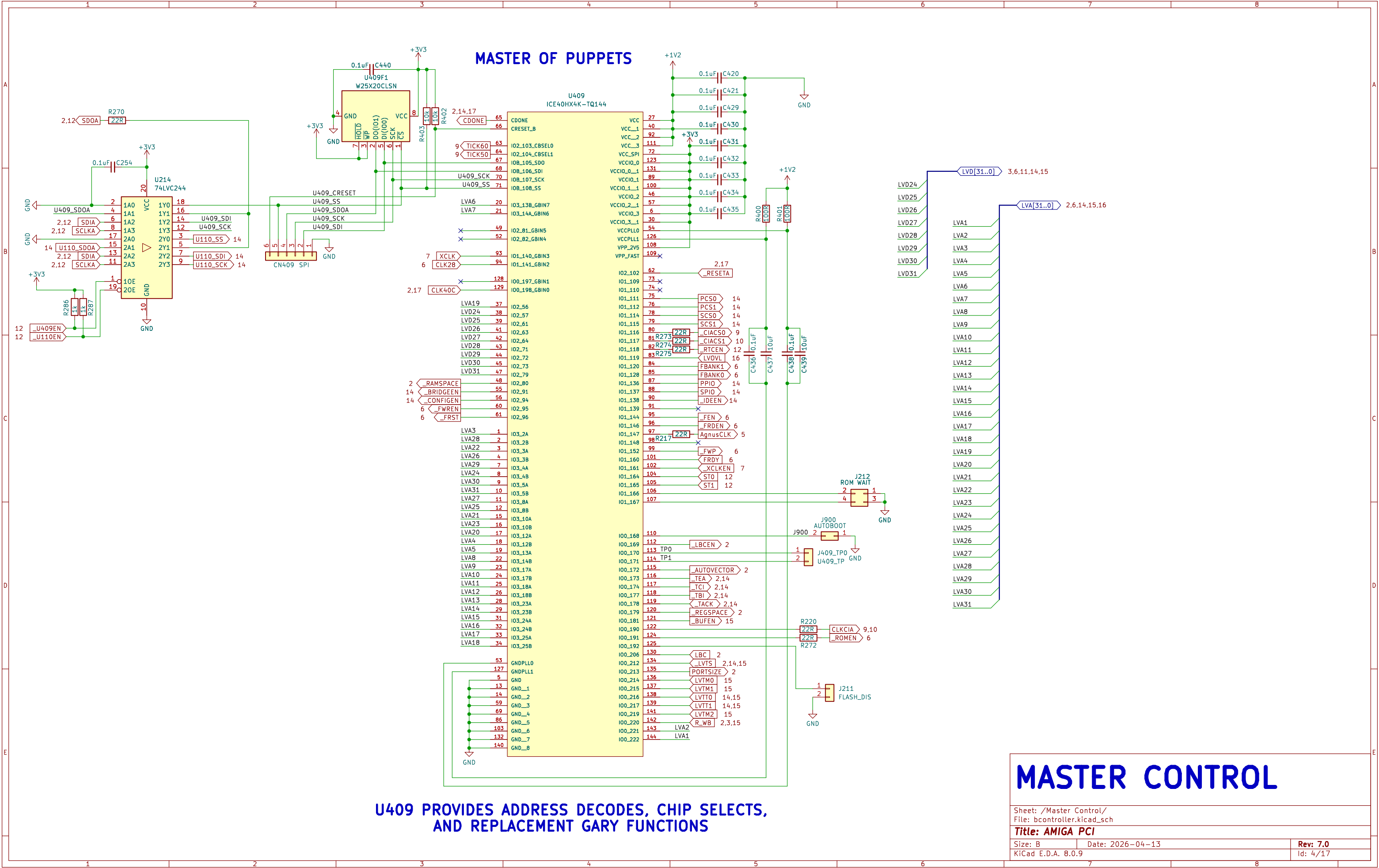
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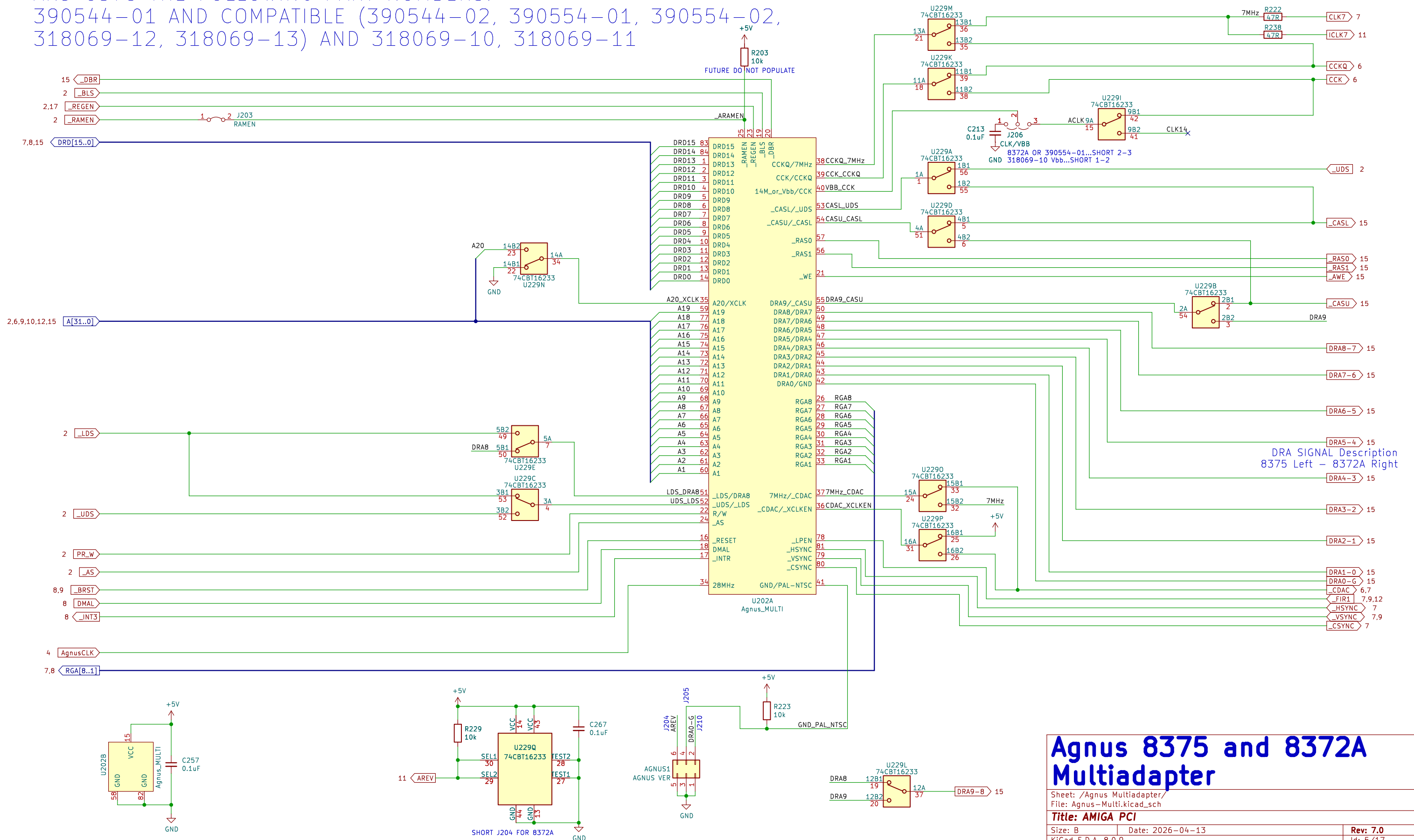


MOST SIGNIFICANT
AGNUS ADDRESSES
00, 01

LEAST SIGNIFICANT
AGNUS ADDRESSES
10, 11

CHIPSET RAM

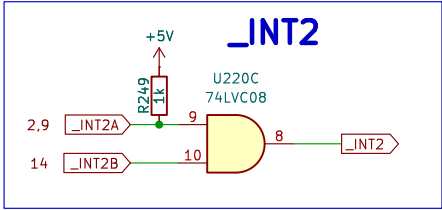
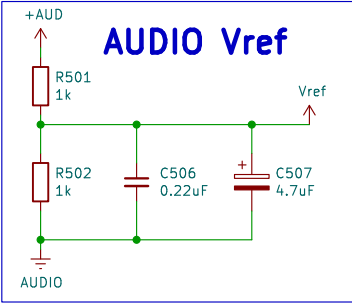
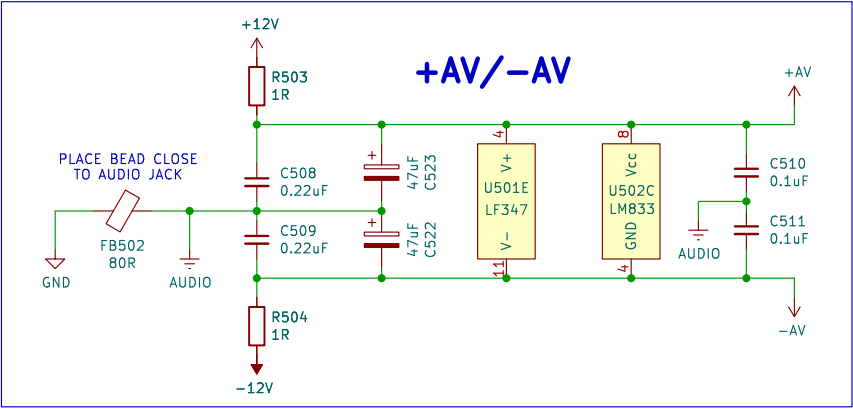
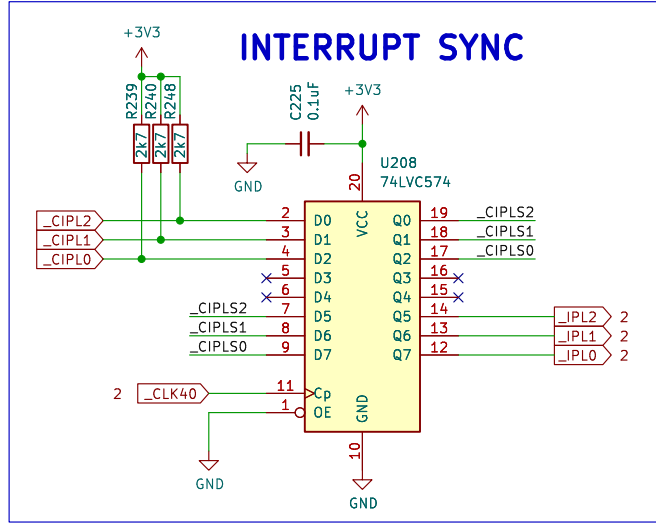
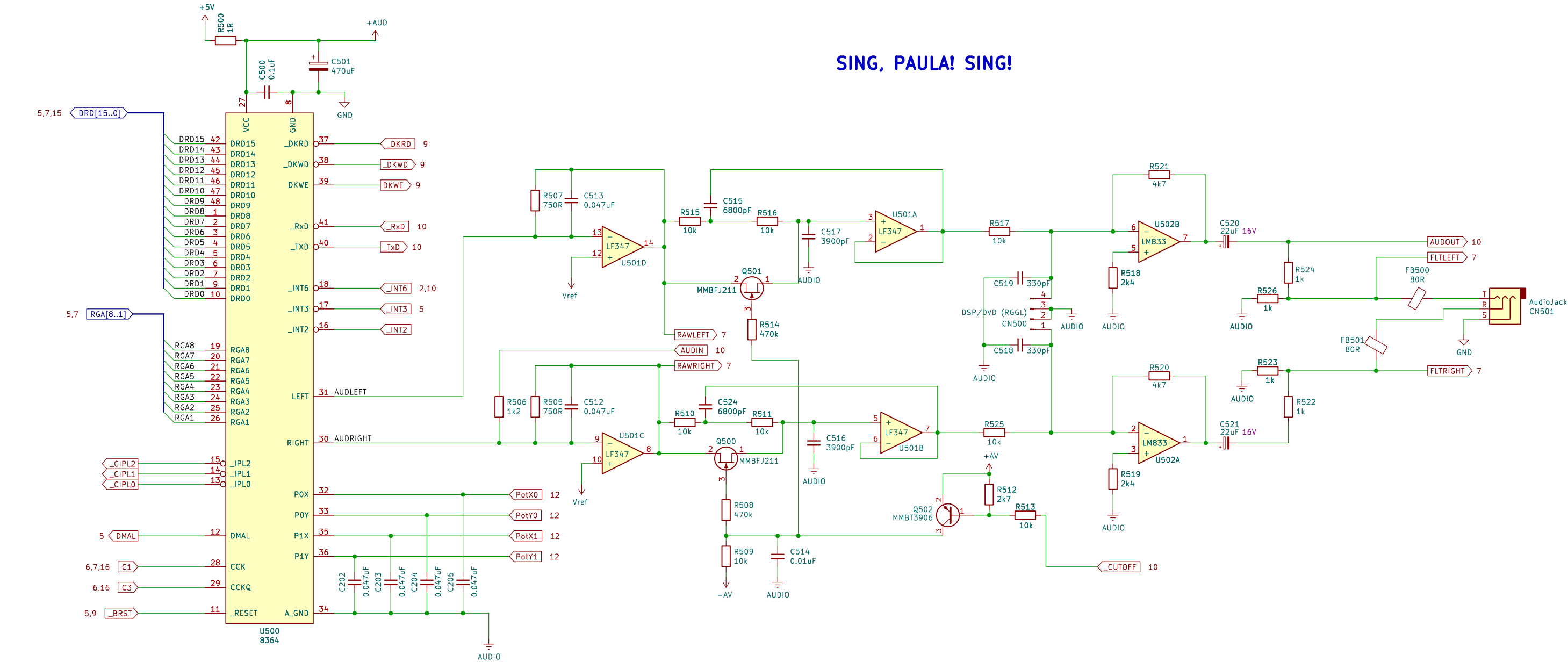


[illegible]

Agnus 8375 and 8372A Multiadapter

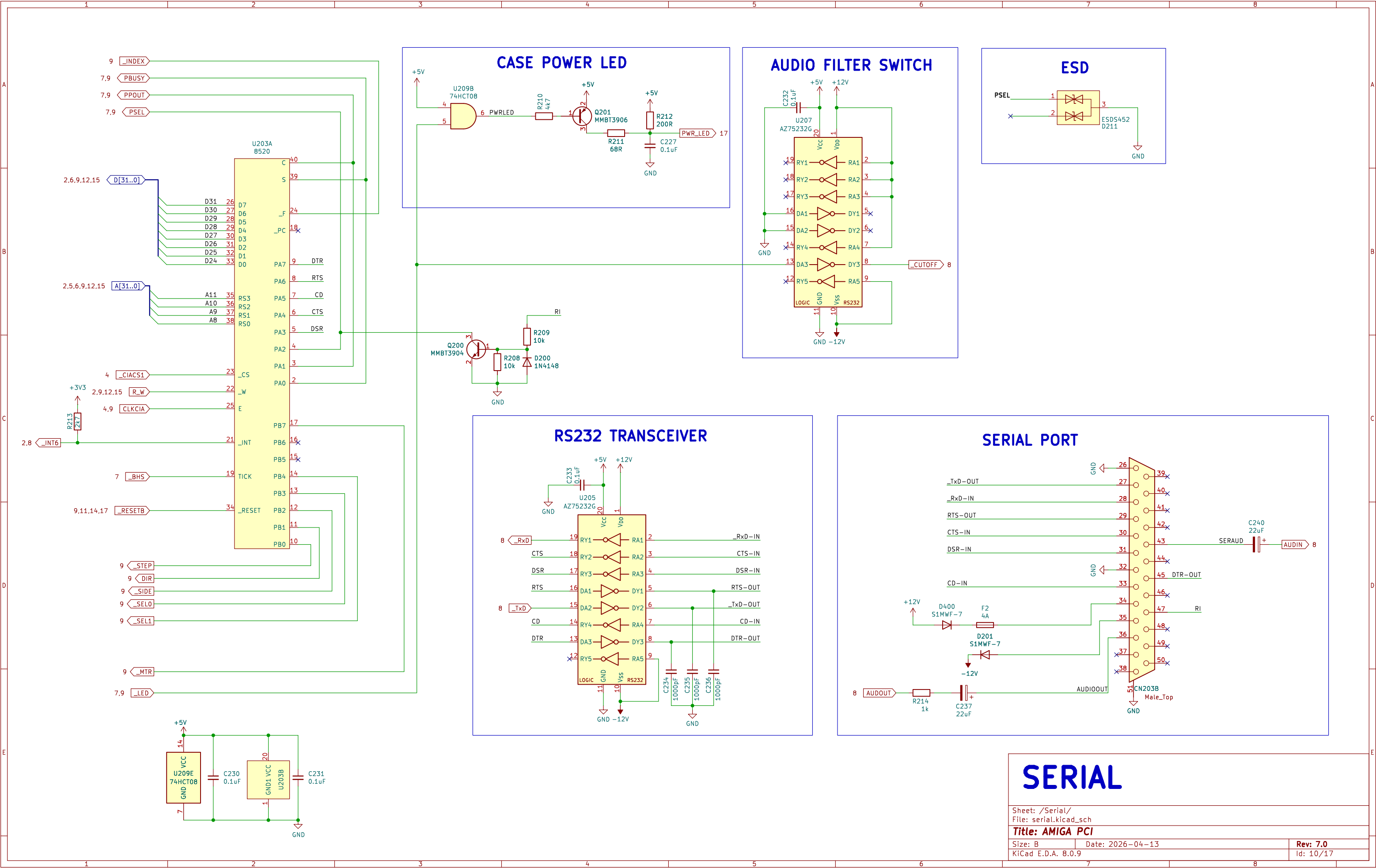
Sheet: /Agnus Multiadapter/		
File: Agnus-Multi.kicad_sch		
Title: AMIGA PCI		
Size: B	Date: 2026-04-13	Rev: 7.0
KiCad E.D.A. 8.0.9		Id: 5/17

SING, PAULA! SING!



PAULA/AUDIO

Sheet: /Audio/ File: audio.kicad_sch		
Title: AMIGA PCI		
Size: B	Date: 2026-04-13	Rev: 7.0
KiCad E.D.A. 8.0.9		Id: 8/17



SERIAL

Sheet: /Serial/ File: serial.kicad_sch		
Title: AMIGA PCI		
Size: B	Date: 2026-04-13	Rev: 7.0
KiCad E.D.A. 8.0.9	Id: 10/17	

74LVC245 ENABLED FOR FIRST WORD DURING WRITES.
DATA ALWAYS FLOWS A->B.

74LVC373 LATCHES DATA FOR FIRST WORD DURING READS.

DURING WRITE CYCLES, ALL DATA IS REAL-TIME.
FOR THE FIRST WRITE COMMAND, D16..31 IS ENABLED.
FOR THE SECOND WRITE COMMAND, D0..15 IS ENABLED.

DURING READ CYCLES, D16..31 IS LATCHED
AFTER THE FIRST READ COMMAND FROM THE IDE DEVICE.
THE SECOND READ FROM THE ATA DEVICE DRIVES
D0..15 LIVE. THE DATA IS THEN LATCHED BY THE MC68040.

DATA BUFFERS

The diagram illustrates the data buffer architecture for the IDE interface. It consists of four 74LVC245 buffers (U900A, U904A, U901A, U905A) and two 74LVC373 latches (U906, U907). The buffers are connected to the IDE bus (LVD[31..0]) and the ATA device (LVD[0..31]). The latches are used to latch data during read cycles. The diagram includes pin connections for A0-A7, B0-B7, IDED0-IDED15, IDEDIR, IDELEN, IDEHVEN, IDEHREN, and power pins (+3V3, GND).

U900A 74LVC245

Pin	Signal	Pin	Signal
2	LVD12	18	IDED12
3	LVD3	17	IDED3
4	LVD13	16	IDED13
5	LVD2	15	IDED2
6	LVD14	14	IDED14
7	LVD1	13	IDED1
8	LVD15	12	IDED15
9	LVD0	11	IDED0

U904A 74LVC245

Pin	Signal	Pin	Signal
2	LVD23	18	IDED7
3	LVD24	17	IDED8
4	LVD22	16	IDED6
5	LVD25	15	IDED9
6	LVD21	14	IDED5
7	LVD26	13	IDED10
8	LVD20	12	IDED4
9	LVD27	11	IDED11

U901A 74LVC245

Pin	Signal	Pin	Signal
2	LVD7	18	IDED7
3	LVD8	17	IDED8
4	LVD6	16	IDED6
5	LVD9	15	IDED9
6	LVD5	14	IDED5
7	LVD10	13	IDED10
8	LVD4	12	IDED4
9	LVD11	11	IDED11

U905A 74LVC245

Pin	Signal	Pin	Signal
2	LVD28	18	IDED12
3	LVD19	17	IDED3
4	LVD29	16	IDED13
5	LVD18	15	IDED2
6	LVD30	14	IDED14
7	LVD17	13	IDED1
8	LVD31	12	IDED15
9	LVD16	11	IDED0

U906 74LVC373

Pin	Signal	Pin	Signal
00	D0	2	LVD23
01	D1	5	LVD24
02	D2	6	LVD22
03	D3	9	LVD25
04	D4	12	LVD21
05	D5	15	LVD26
06	D6	16	LVD20
07	D7	19	LVD27

U907 74LVC373

Pin	Signal	Pin	Signal
00	D0	2	LVD28
01	D1	5	LVD19
02	D2	6	LVD29
03	D3	9	LVD18
04	D4	12	LVD30
05	D5	15	LVD17
06	D6	16	LVD31
07	D7	19	LVD16

BUFFERS AND TERMINATION

The image displays two circuit diagrams for 74LVC245 chips, U902A and U903A, showing their pin connections and termination configurations.

U902A 74LVC245:

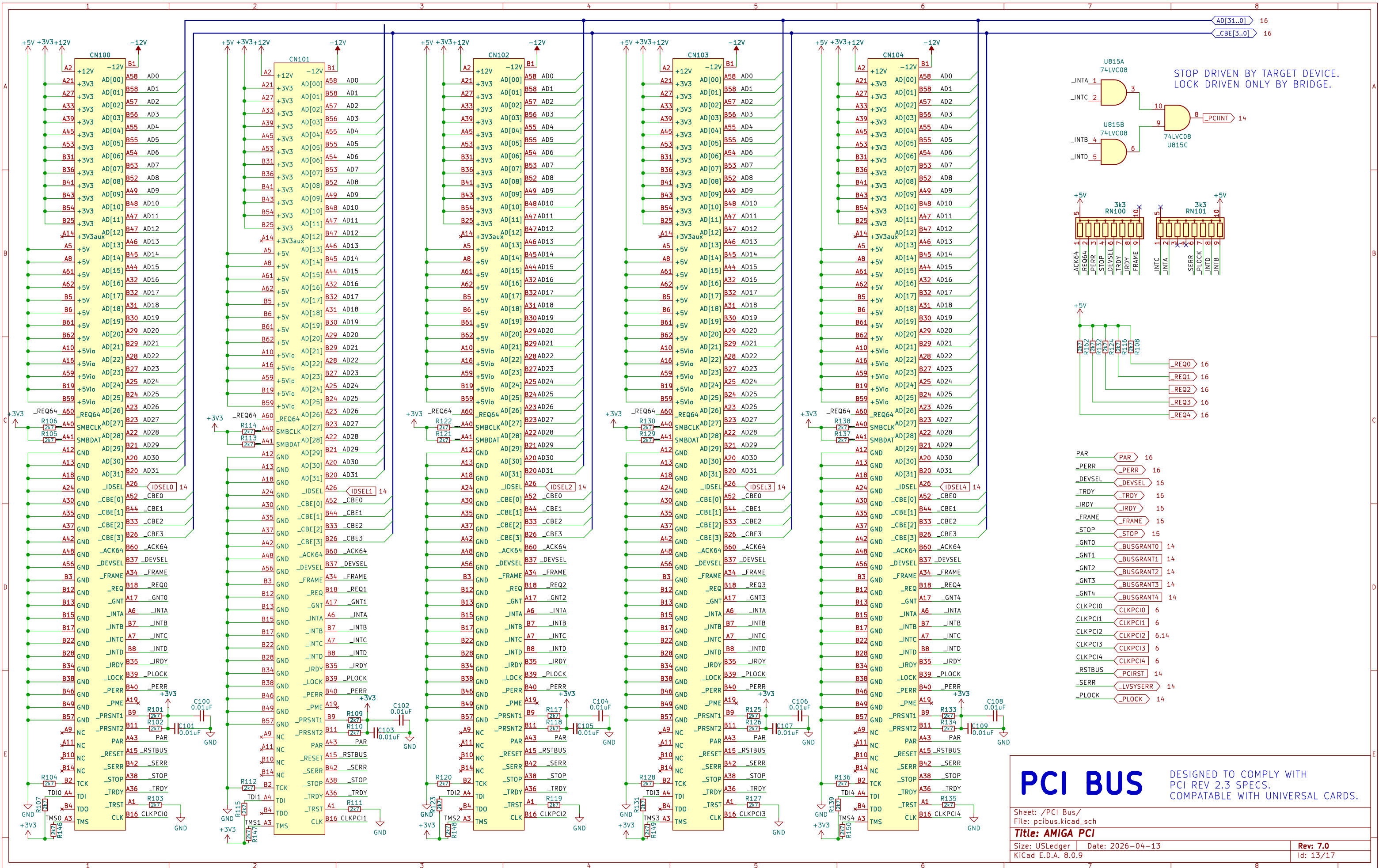
- Inputs (Left):**
 - DA1B (18) to RN906A (1)
 - DA0B (17) to RN906B (2)
 - DA2B (16) to RN906C (3)
 - INTRP (14) to INTRP (14)
 - INTRS (13) to INTRS (13)
 - AGNUS_REV (12) to AGNUS_REV (12)
 - LVCLK7 (11) to LVCLK7 (11)
- Outputs (Right):**
 - DA1 (2) to DA1 (2)
 - DA0 (2) to DA0 (2)
 - DA2 (2) to DA2 (2)
 - INTRQP (5) to INTRQP (5)
 - INTRQS (7) to INTRQS (7)
 - AREV (8) to AREV (8)
 - ICLK7 (9) to ICLK7 (9)
- Internal Connections:**
 - DA1A (18) to DA1A (18)
 - DA0A (17) to DA0A (17)
 - DA2A (16) to DA2A (16)
 - RSTIDE (15) to RSTIDE (15)
 - R215 (11) to R215 (11)
- Power and Ground:**
 - +3V3 (1) to +3V3 (1)
 - GND (19) to GND (19)

U903A 74LVC245:

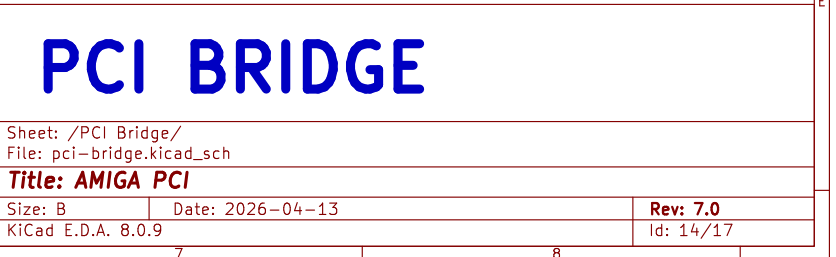
- Inputs (Left):**
 - DIOWS (18) to RN905A (1)
 - DIORS (17) to RN905B (2)
 - CS1S (16) to RN905C (3)
 - CS0S (15) to RN905D (4)
 - DIOWP (14) to RN904A (5)
 - DIORP (13) to RN904B (6)
 - CS0P (12) to RN904C (7)
 - CS1P (11) to RN904D (8)
- Outputs (Right):**
 - DIOWB (14) to DIOWB (14)
 - DIORB (14) to DIORB (14)
 - CS1B (14) to CS1B (14)
 - CS0B (14) to CS0B (14)
 - DIOWA (14) to DIOWA (14)
 - DIORA (14) to DIORA (14)
 - CS0A (14) to CS0A (14)
 - CS1A (14) to CS1A (14)
- Internal Connections:**
 - DIOWA (14) to DIOWA (14)
 - DIORB (14) to DIORB (14)
 - CS1B (14) to CS1B (14)
 - CS0B (14) to CS0B (14)
 - DIOWA (14) to DIOWA (14)
 - DIORA (14) to DIORA (14)
 - CS0A (14) to CS0A (14)
 - CS1A (14) to CS1A (14)
- Power and Ground:**
 - +3V3 (1) to +3V3 (1)
 - GND (19) to GND (19)



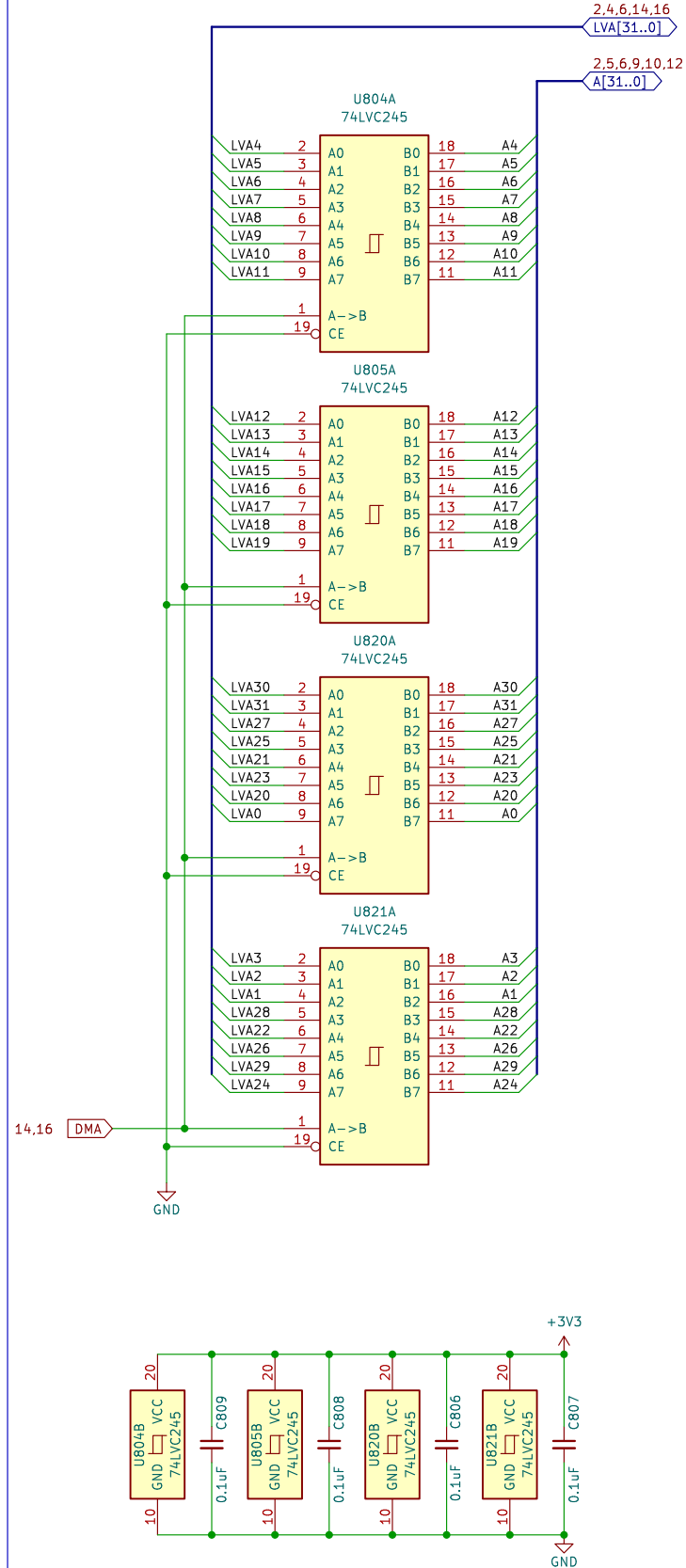
Size: B	Date: 2026-04-13	Rev: 7.0
KiCad E.D.A. 8.0.9		Id: 11/17



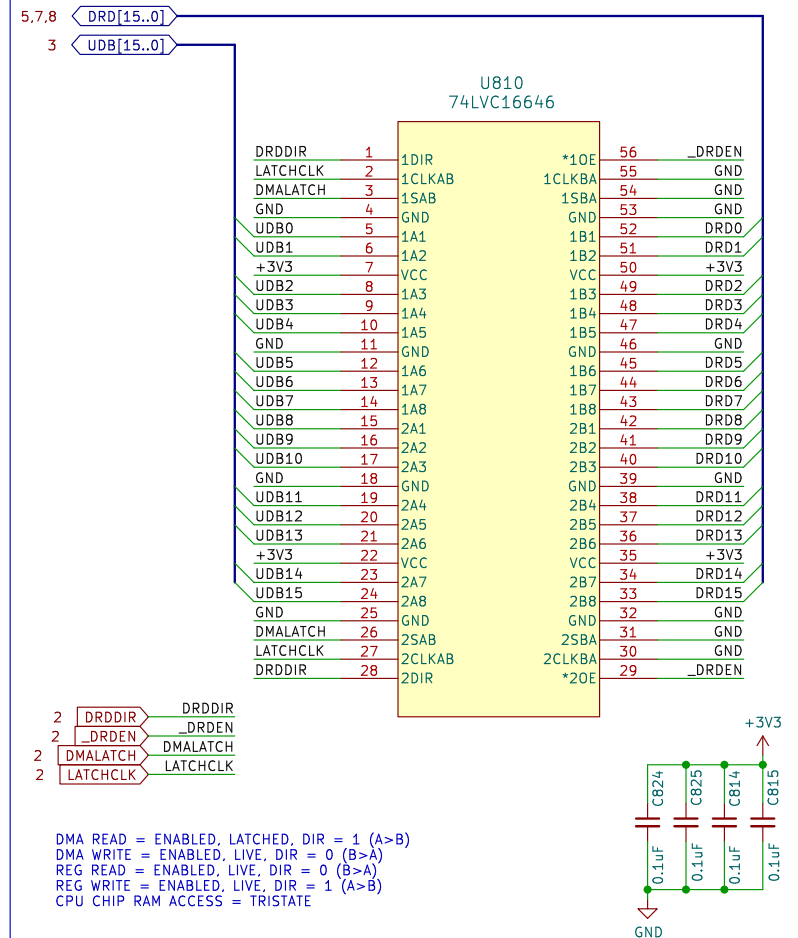
U110: BUS ARBITOR, INTERRUPT HANDLING, PARITY ERROR DETECTION, ATA CONTROLLER



LOW VOLTAGE ADDRESS BUFFERS



LOW VOLTAGE CHIPSET RAM DATA BUS BUFFER/LATCH

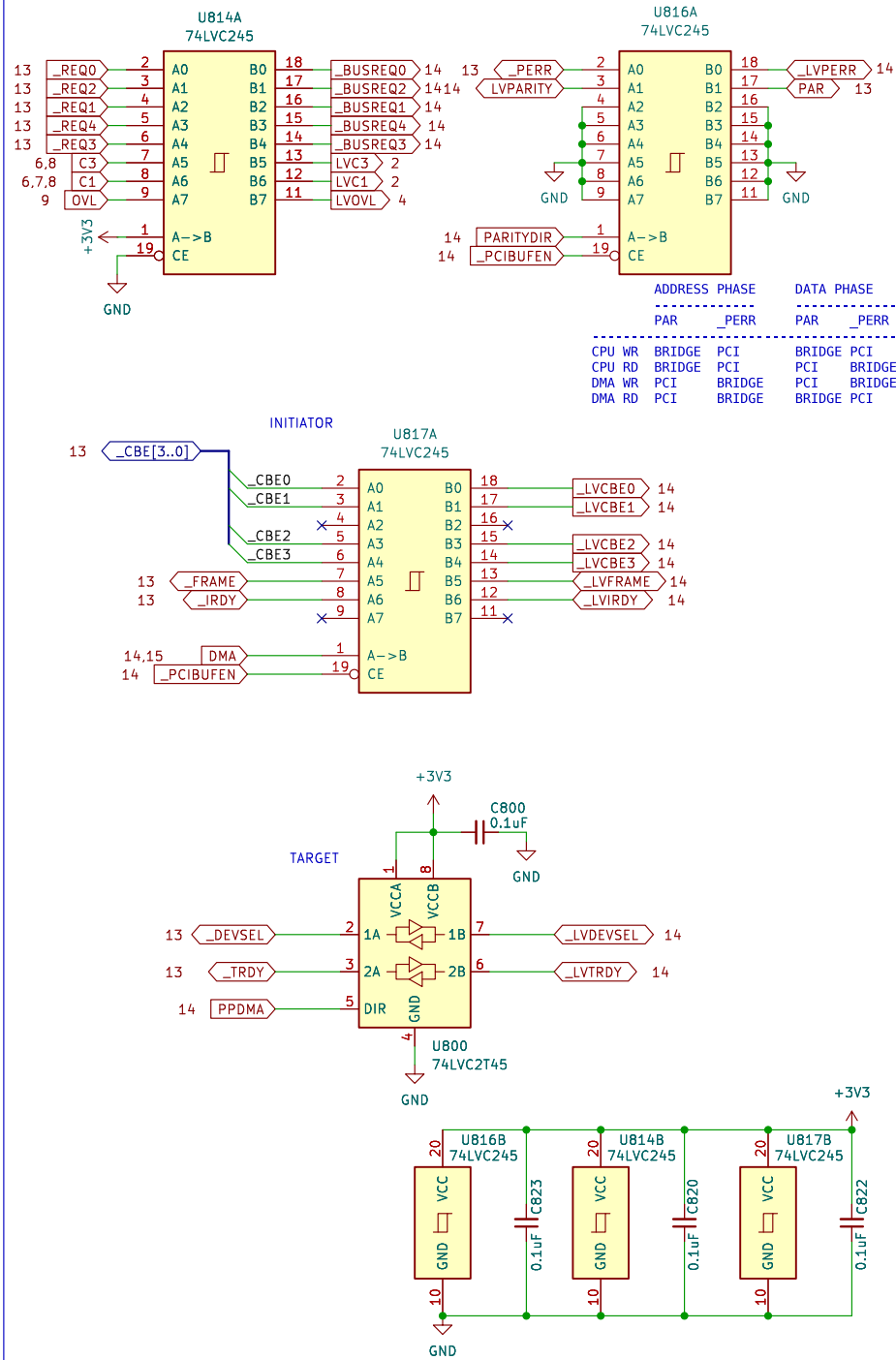


The diagram shows two 74LVC245 buffers, U808A and U809A, connected in a chain. The U808A is connected to the U809A. The U808A has inputs LVSIZ1, LVTM0, R_WB, R_WC, LVTM1, LVTTO, LVTI1, LVTM2, DMA, and CE. The U809A has inputs LVCPULOCK, LVSTOP, LVTS, LVSIZ0, LVBB, and CE. Both buffers have outputs SI21, TM0, R_W, TM1, TIO, TI1, TM2, and CE. The buffers are connected to a 3V3 supply and ground through capacitors C812 and C813.

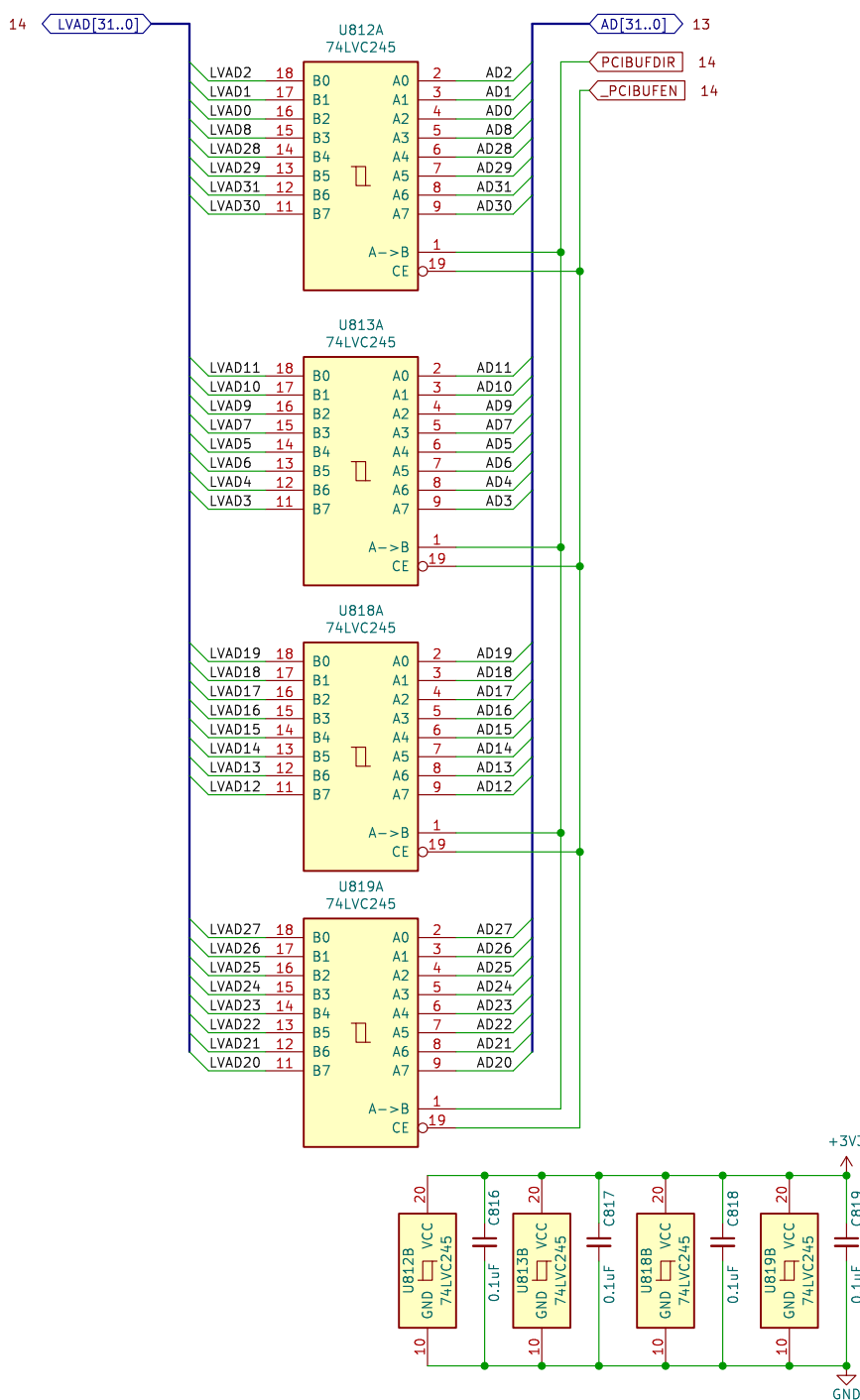
MOTOROLA BUS LOGIC

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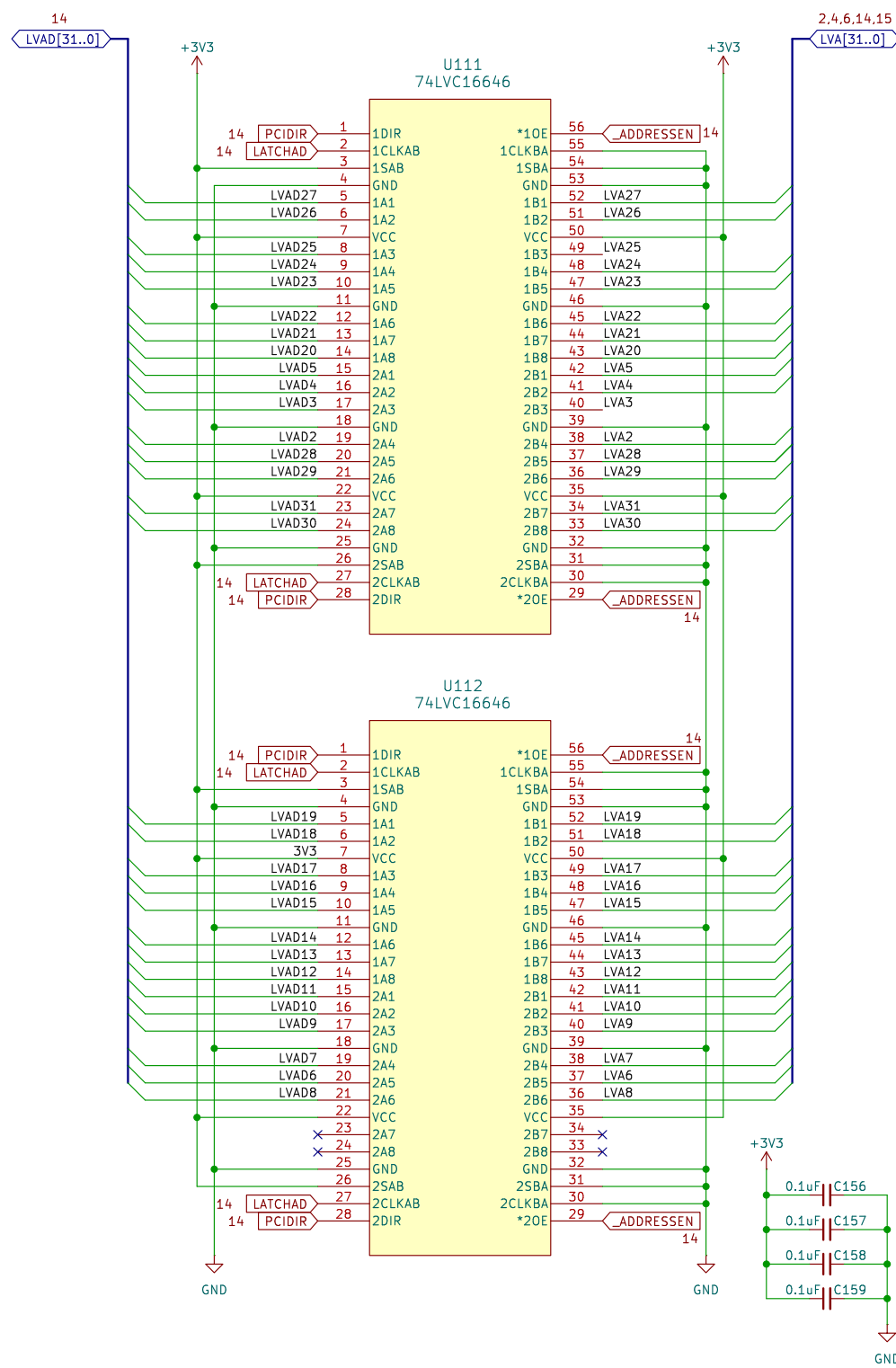
PCI BUFFERS

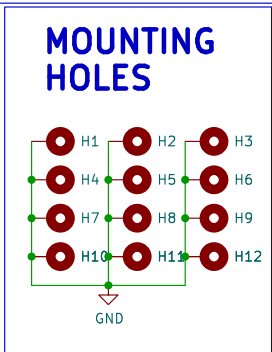
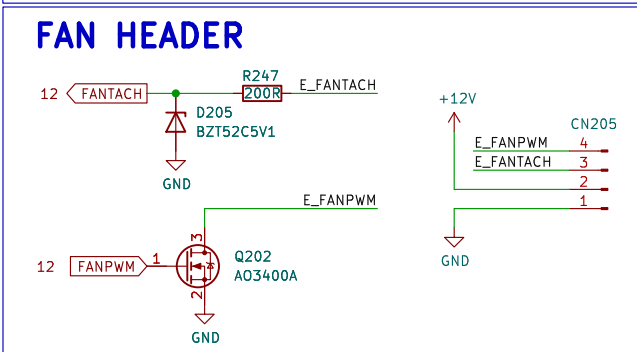
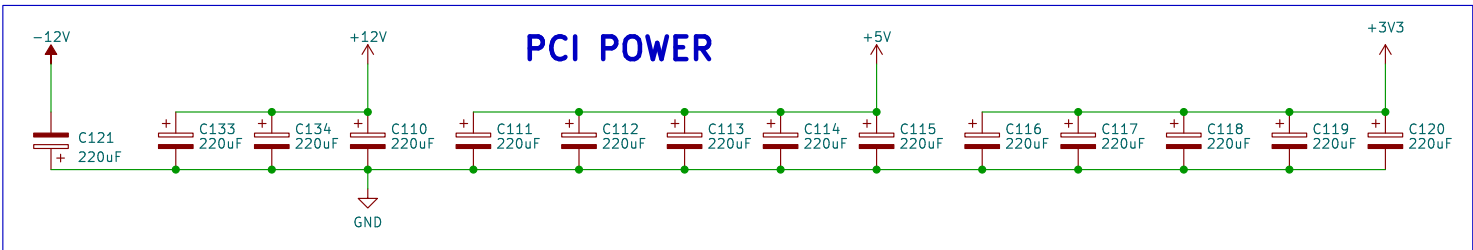
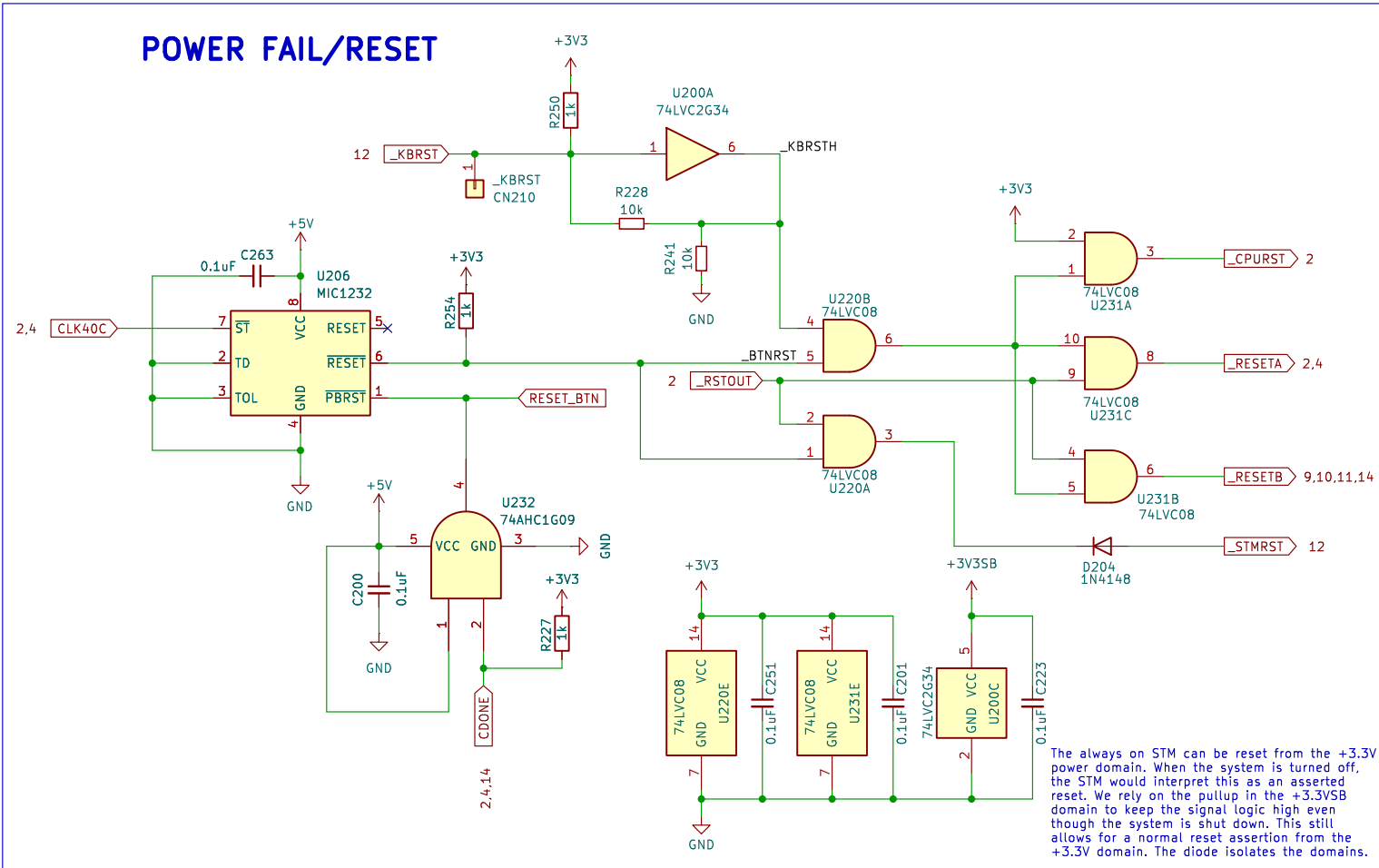
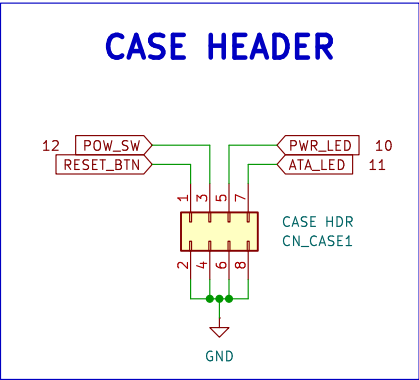
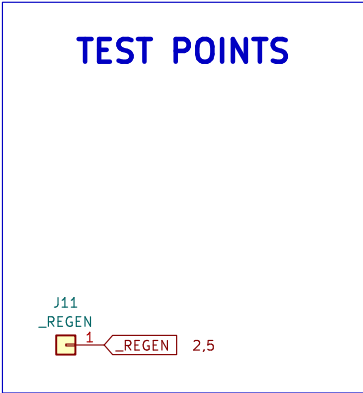
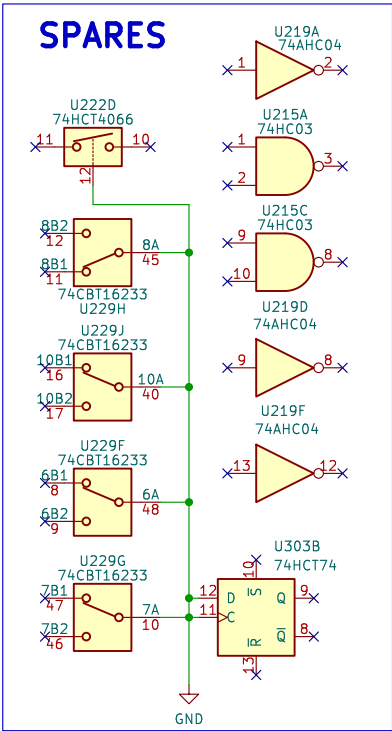
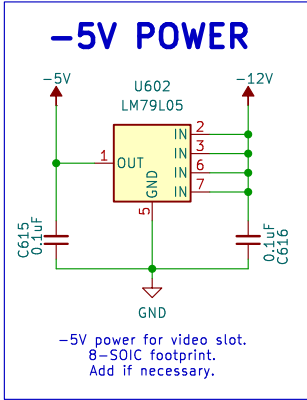
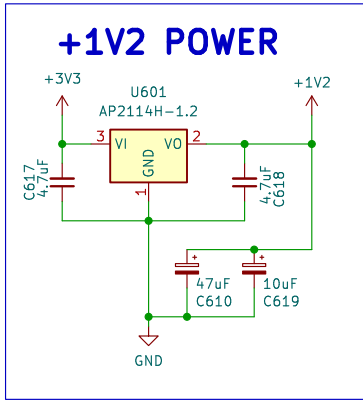
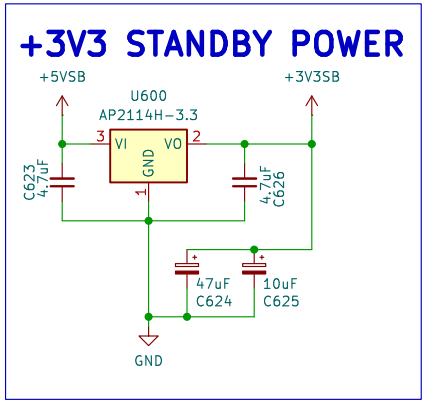
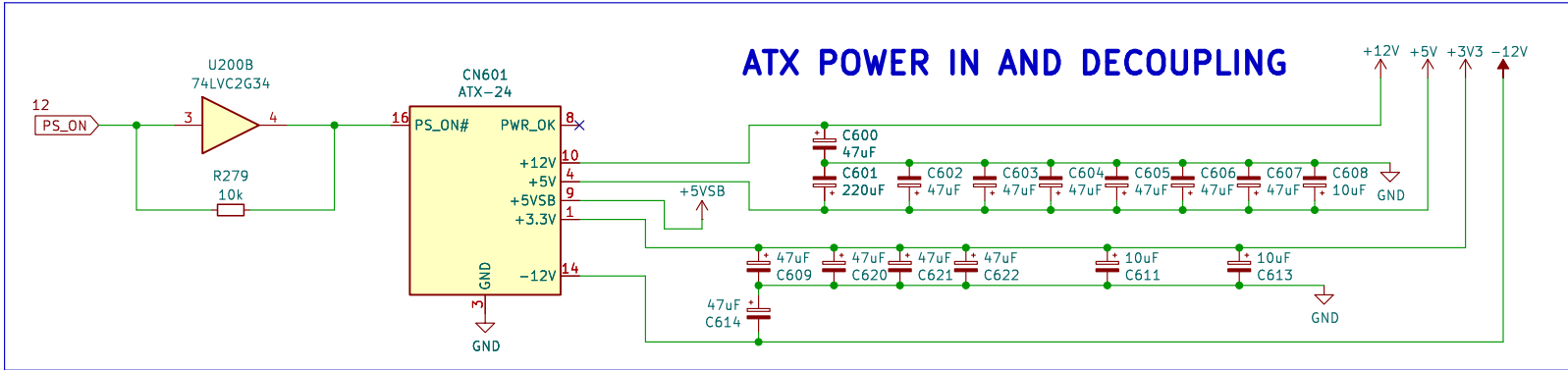


AD BUS LEVEL SHIFT



PCI ADDRESS LATCH





POWER/SPARES

Sheet: /Power and Spares/ File: power.kicad_sch		
Title: AMIGA PCI		
Size: B	Date: 2026-04-13	Rev: 7.0
KiCad E.D.A. 8.0.9	Id: 17/17	